



[3. Resizing, Interpolation and](#)
[Course](#) > [Mathematical Operations](#) > [3.4. Conclusion](#) > 3.4.2. Weekly Graded Assessment

3.4.2. Weekly Graded Assessment

Interpolation / Resizing

4.0/4.0 points (graded)

When performing image processing and analysis, it is important to match your sampling rate (pixel size) to your feature size (the object you want to measure). Why?

Check all that apply

- ☒ you need your object to be defined by enough pixels to be able to extract meaningful shape information ✓
- ☒ oversampling your feature will not bring more information but might make processing slower (more images or larger images) ✓
- ☒ enlarging your image digitally (scaling) does not bring new information. It must be done at acquisition ✓
- ☒ if you acquired an oversampled image, you can scale it down before processing to gain time. But you've acquired more data than necessary. ✓
- ☒ matching sampling rate to object size is an application of the Shannon-Nyquist theorem ✓



Submit

i Answers are displayed within the problem

Proper Feature Size

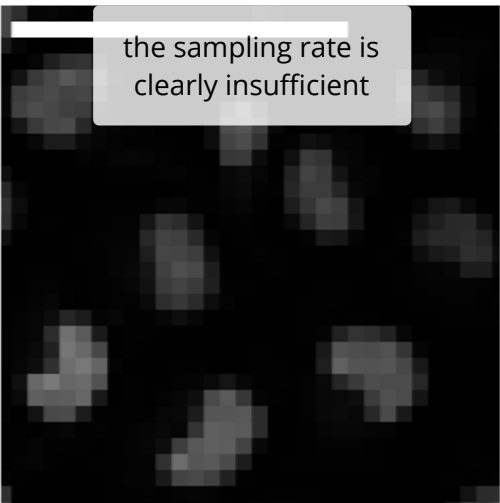
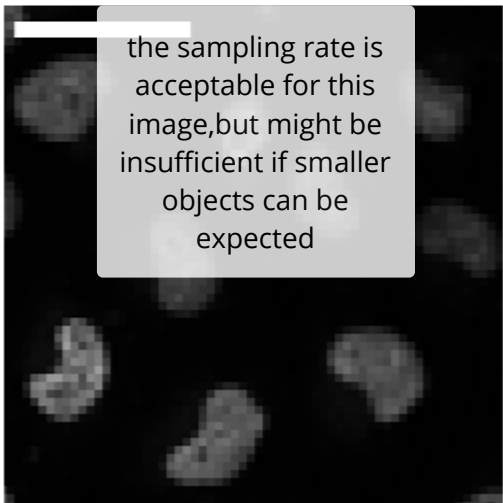
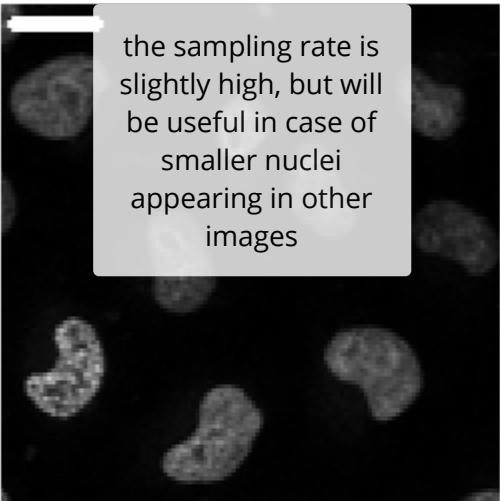
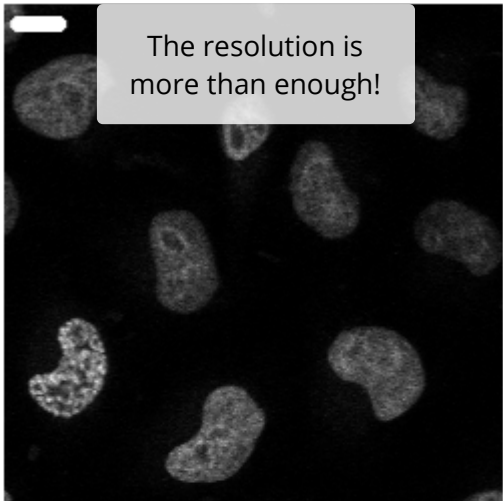
2/2 points (graded)

Keyboard Help

PROBLEM



A point scanning confocal microscope allows you to select your sampling rate (pixel size). Someone has taken images of DAPI stained cells with multiple sampling rates and wants to measure the nuclear area.
Drag the comments on the adequate image.
The white bars on the top left represent 20 pixels on each image



Submit

You have used 2 of 2 attempts.

Reset

Show Answer

FEEDBACK

✓ Correctly placed 4 items.

✓ Always think of what you need to image and adapt your experimental conditions to acquire enough (but not too much data). You should never do this per image of your experiment. Once a proper sampling rate has been found, you must keep it constant throughout your experiment.

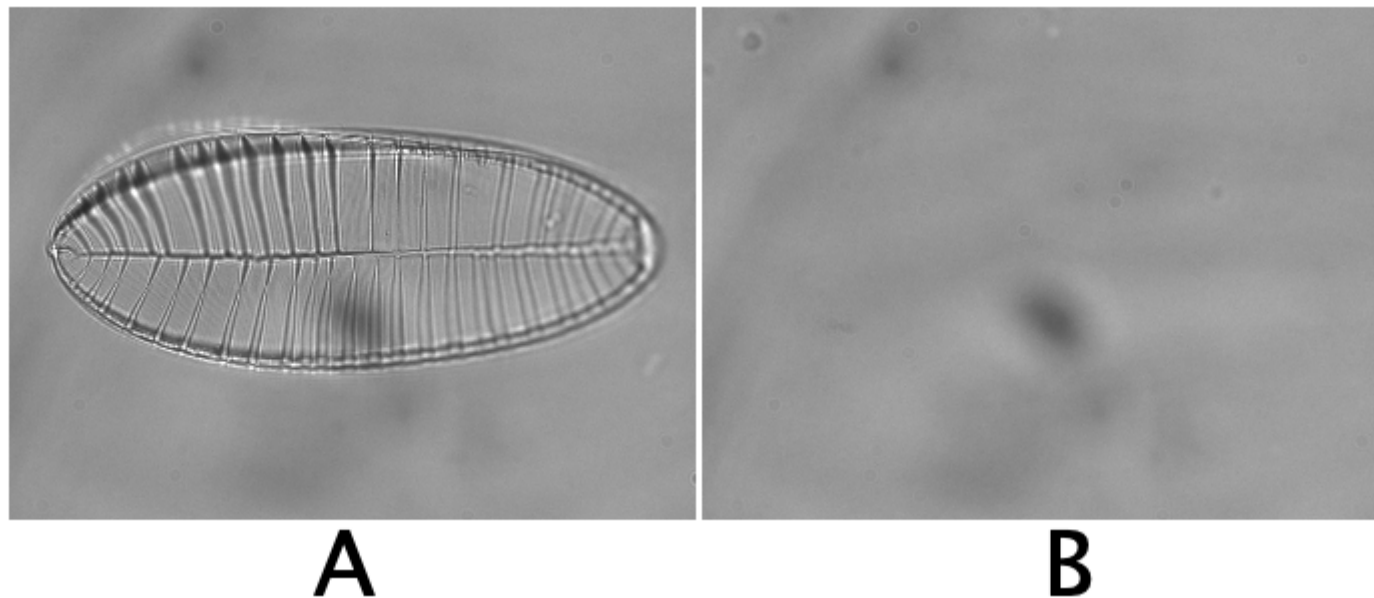
✓ Final attempt was used, highest score is 2.0

Image to Image Math

2.0/2.0 points (graded)

You have acquired a brightfield image of a diatom and have found that the image has artifacts **(A)**.

To check that it is not your sample, you decide to acquire an image with the sample removed. You notice that the same artifacts appear whether or not the sample is there **(B)**.



You can find the corresponding images in [this week's downloads](#) (*WF Diatom.tif* and *WF Empty Field.tif*) and try for yourself.

What image-to-image operation can you perform to compensate for the uneven background?

☐ Dividing B with A

☒ Subtracting B to A

☒ Dividing A with B

☐ Adding B to A

☐ Multiplying A and B

☐ Subtracting A to B



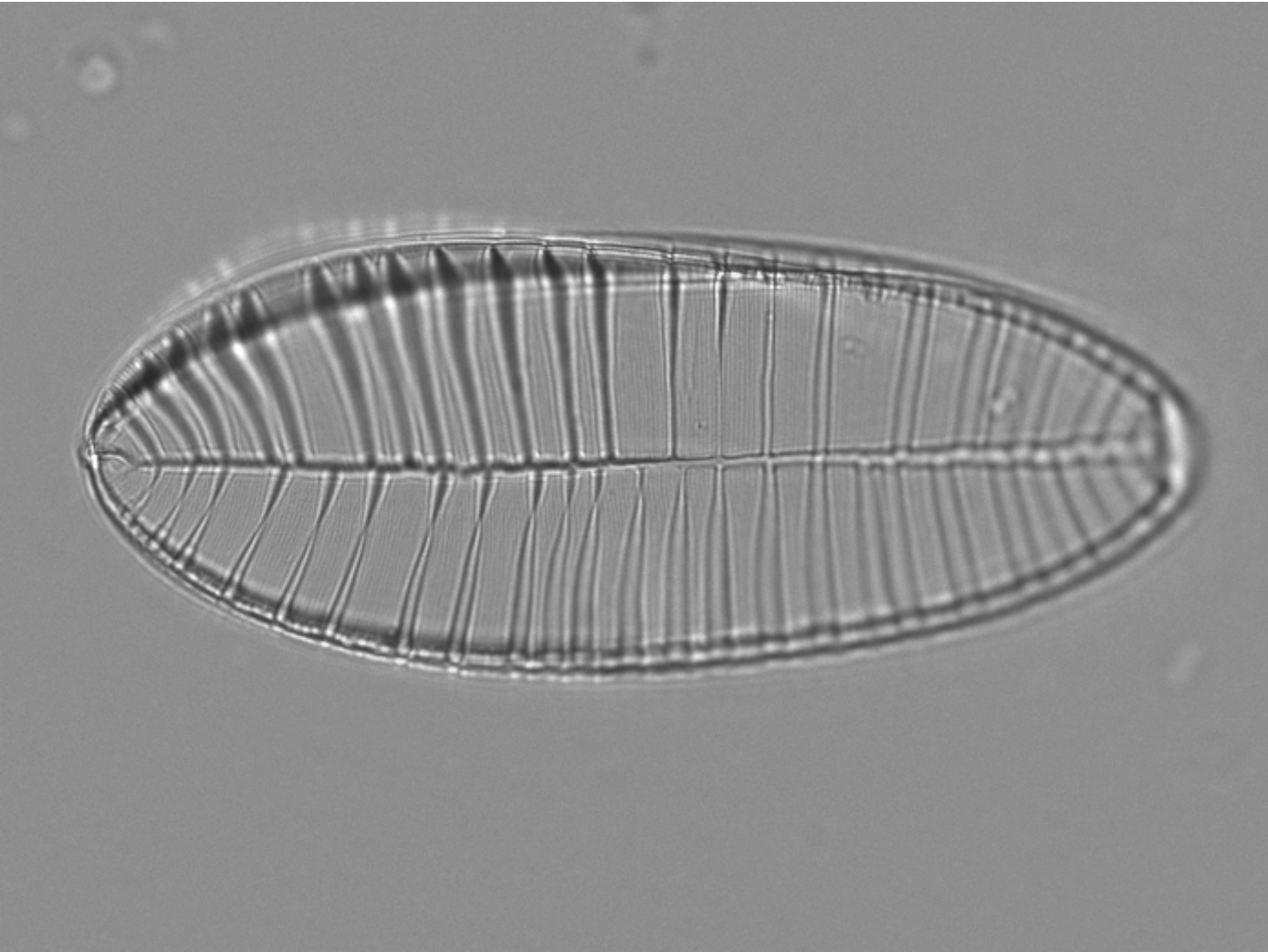
Submit

You have used 1 of 2 attempts

Image Math

2.0/2.0 points (graded)

The resulting image **R** from the previous question is given here as a reference



We want to perform the same kind of correction, but we want to normalize the intensities of the resulting image **R** so that it has the same mean as the original diatom image **A**.

What is the order of steps we should perform in ImageJ to do this?

(Assume we know the mean intensity values of **A** and **R**)

Not all steps are useful or correct

1. Divide **R** by its mean intensity for each pixel
2. Divide **A** with **R**, setting the resulting image to 32-bits
3. Subtract the mean of **A** to **R** for each pixel
4. Multiply **R** by the mean intensity of **A** for each pixel
5. Make sure **R** is set to 32-bits
6. Divide **A** by the mean intensity of **R** for each pixel

Enter the order of the operations to perform, separating each one with a dash '-' (eg. 1-2-3)

☐ 1--4--2--3

☐ 2--5--6

☐ 6--5--4

☒ 5--1--4 ✓

☐ 3--4--6--1

Submit

You have used 1 of 2 attempts

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